

Summary Report

EV Adoption & Trends Report

Objective

To analyze Electric Vehicle (EV) adoption trends by model year, range performance, and geographic distribution to guide policy, infrastructure planning, and business strategy.

Key Findings & KPIs

1. EV Registrations Over Time

 Strong growth post-2017, peaking in 2023 with ~60,000 registrations.

 Drop after 2023 (-17% in 2024, -41% in 2025) – signals potential policy or supply constraints.

YoY growth exceeded +100% in 2023, showing market acceleration.

KPI: 2023 = Peak adoption year with 103% growth YoY.

2. Electric Range by EV Type

BEVs (Battery EVs): Median = 215 mi, widespread (100–300+ mi).

PHEVs (Plug-in Hybrids): Median = 25 mi, tightly clustered around short ranges.

Clear technology gap: BEVs dominate in range and long-term potential.

KPI: BEV median range ~9× higher than PHEV.

3. Range Evolution Over Model Years

BEV ranges improved +136% since early 2000s.

PHEV ranges remain relatively flat (slight decline).

Innovation is concentrated in BEVs.

KPI: BEVs reached 260+ miles avg in 2020, PHEVs stagnant ~25–40 mi.

4. Geographic Adoption – Counties

King County dominates ~50% of all EVs in the state.

Top 4 counties (King, Snohomish, Pierce, Clark) = 76% of statewide EVs.

EVs are still highly concentrated in urban/suburban **counties**.

KPI: King County = 128,000+ EVs (~50% of total).

5. Geographic Adoption – Cities

Seattle alone has 41,000+ EVs (15.8%) – far ahead of others.

Next leaders: Bellevue (4.9%), Vancouver (3.7%), Redmond (3.4%), Bothell (3.3%).

Top 5 cities account for ~31% of statewide EVs.

KPI: Seattle's EV count $\approx 3.5 \times$ Bellevue's, showing its role as the EV epicenter.

Strategic Insights

Seattle & King County = Core EV Hub

Prioritize charging infra, dealerships, and incentives here first.

Secondary Growth Cities

Bellevue, Vancouver, Redmond are high-potential suburban clusters.

Technology Shift → BEVs dominate

PHEVs are transitional; long-term infra & policy should focus on BEVs.

Market Risks

Post-2023 decline in registrations suggests need for sustained policy support, affordability initiatives, and supply chain stability.

Summary KPIs

 2023 Peak Registrations: 60,000 EVs (+103% YoY).

 Median Range: BEV = 215 mi | PHEV = 25 mi.

 King County Share: 50% of all EVs.

 Seattle Share: 41,000 EVs (15.8% of total).

 BEV Range Growth: +136% since early 2000s



Business Problem

Electric Vehicle (EV) adoption in Washington State surged over the past decade, peaking in 2023 with record registrations. However, subsequent declines raise concerns around sustainability, market stability, and infrastructure readiness.



Analytical Scope

- **Highlights technology preferences and consumer adoption trends (BEVs vs PHEVs).**
- **Reveals market share concentration and identifies key automakers driving adoption.**
- **Most popular EV models with manufacturers to support service and parts planning.**
- **Quantifies registration growth trends with year-over-year percentage changes.**
- **Compares electric range performance between BEVs and PHEVs.**
- **Benchmarks EV performance using median range across all models.**
- **Analyzes variability and outliers in electric range by vehicle type.**
- **Tracks yearly improvements in average electric range and growth trends.**
- **Identifies top 15 counties with highest EV adoption (share of total).**
- **Breaks down top 20 cities by EV count and contribution to overall adoption.**
- **Consolidates adoption patterns between top 10 cities + “Others” for market clarity.**

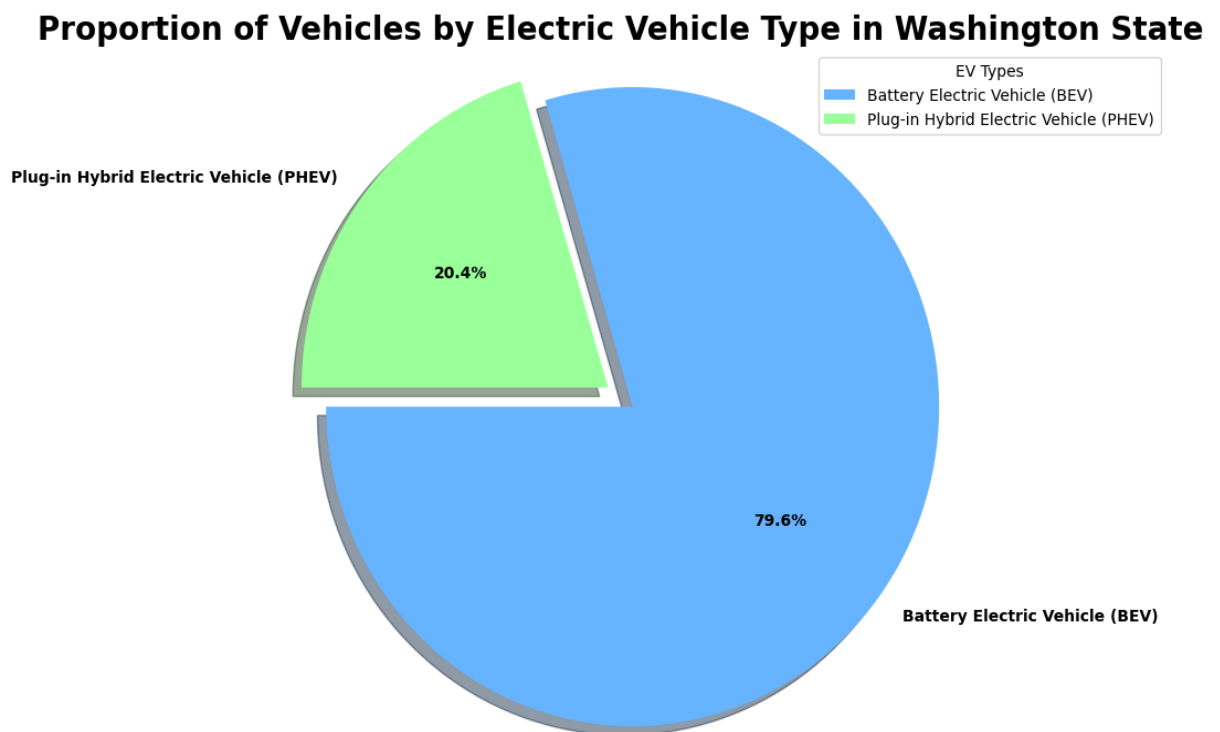
Research Questions & Key Findings: -

1. Preferences and consumer adoption trends (BEVs vs PHEVs).

BEVs dominate the market (79.6%), showing that most Washington residents prefer fully electric vehicles.

PHEVs account for 20.4%, indicating a smaller but still significant segment of consumers who prefer hybrid flexibility.

This distribution suggests strong consumer confidence in full EV adoption in Washington compared to transitional hybrid models.



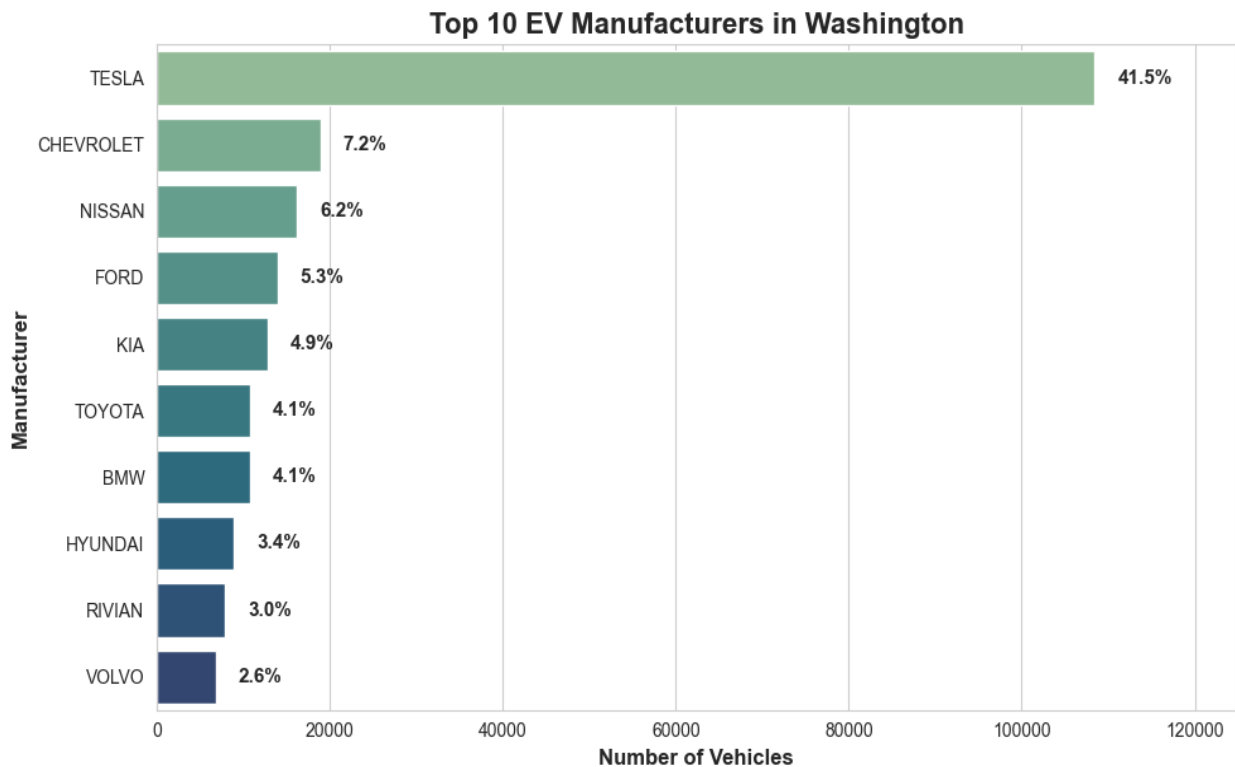
2. Top EV Manufacturers in Washington. DC

Tesla leads by a wide margin (41.5%), making it the dominant EV brand in Washington.

The next closest manufacturers are Chevrolet (7.2%), Nissan (6.2%), and Ford (5.3%). Together they make up less than half of Tesla's share.

The presence of both legacy automakers (Ford, Toyota, BMW, Hyundai, Volvo) and new entrants (Tesla, Rivian) indicates a competitive but uneven market.

The long tail of smaller manufacturers suggests consumer choice diversification is still developing.



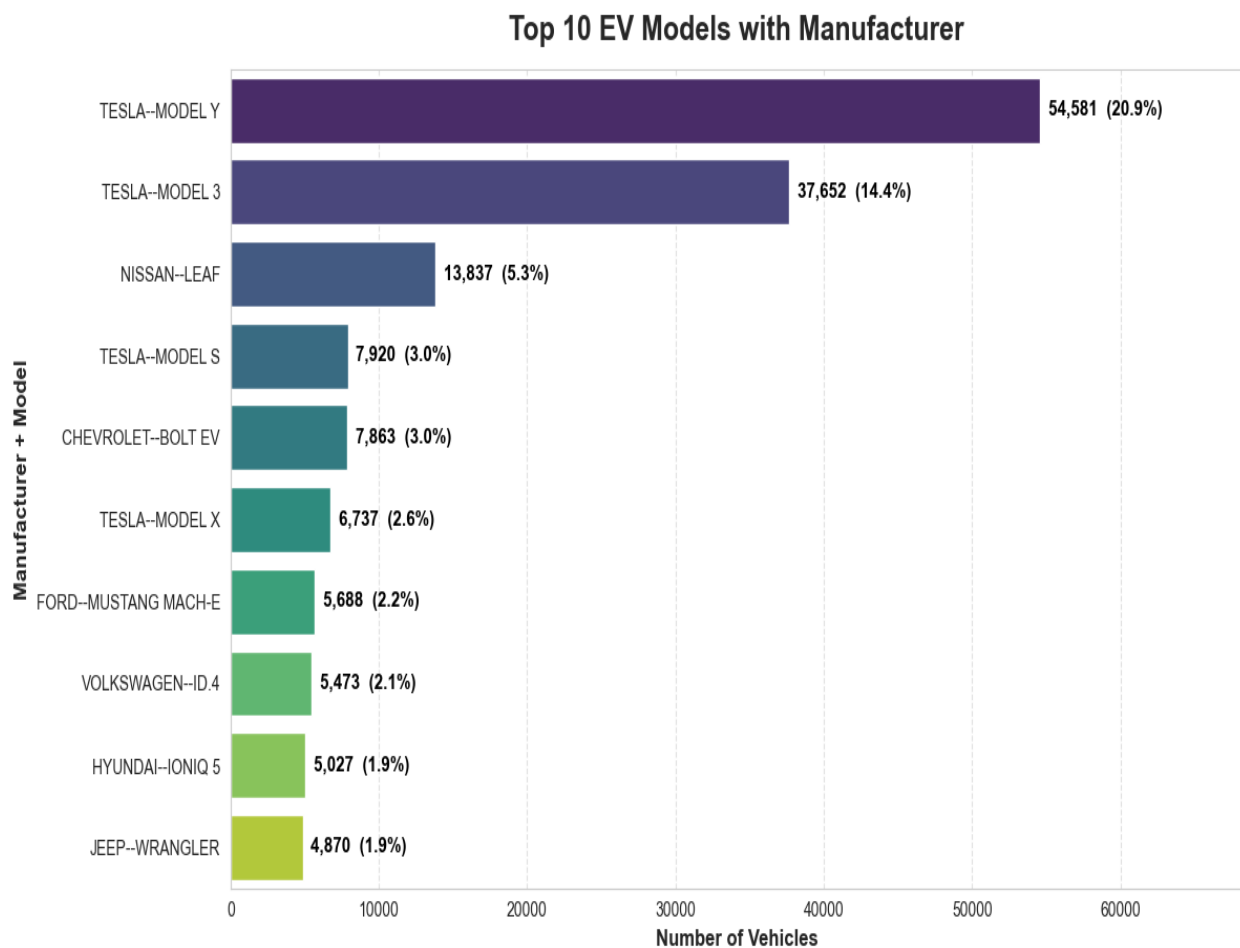
3. Highest EV Models with Manufacturer.

Tesla Model Y (20.9%) and Model 3 (14.4%) dominate, together accounting for more than one-third of all EVs in Washington.

The Nissan Leaf (5.3%) remains a strong contender, showing consumer interest in affordable, practical EV options. While other Tesla models (S and X) also rank high, confirming Tesla's multi-segment appeal (luxury + mainstream).

Chevrolet Bolt EV, Ford Mustang Mach-E, Volkswagen ID.4, Hyundai Ioniq 5, and Jeep Wrangler make up a diverse set of alternatives — but each individually has only ~2–3% share.

This suggests a concentrated market around Tesla, with smaller contributions from legacy brands experimenting with EV models.



4. ⚡ EV Adoption Insights (2000–2026, Washington State)

Early Stage (2000–2010): Very low volumes, highly volatile YoY % (up to +1900%) but not meaningful due to tiny base → reflects experimental/innovation phase.

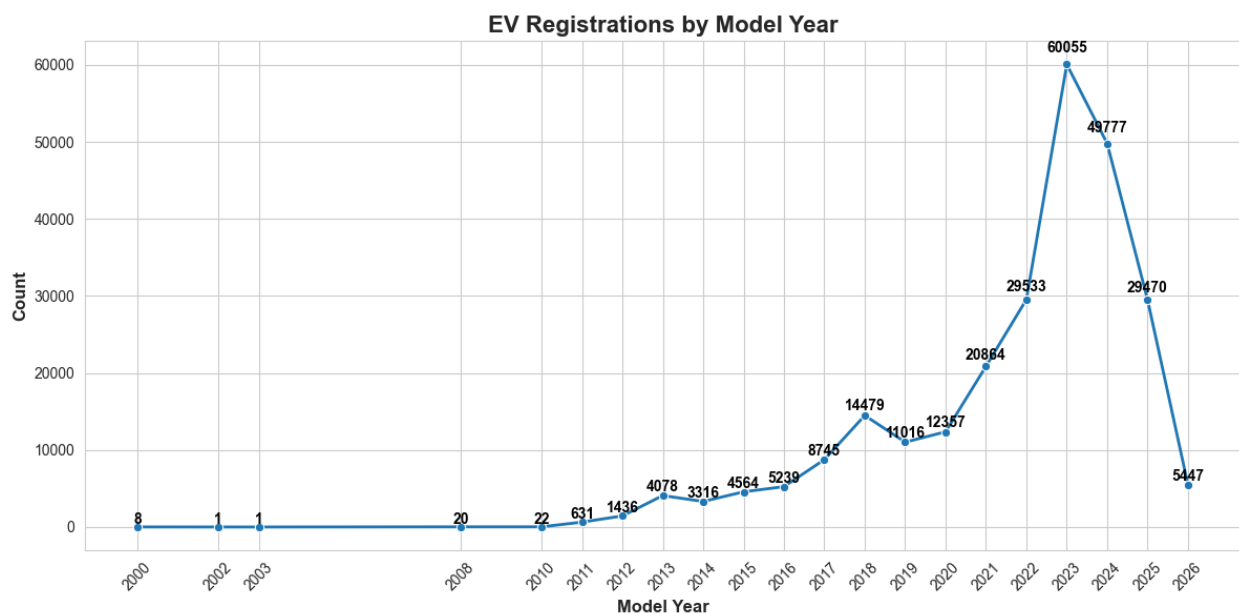
Emerging Growth (2011–2016): Gradual but consistent rise (+30% to +180% YoY) as mainstream EVs (Leaf, Volt) entered market → steady momentum.

Tipping Point (2017–2020): Sharp adoption growth (+66%, +65% in 2017–18), with first signs of scale; slight dip in 2019 (-23.9%).

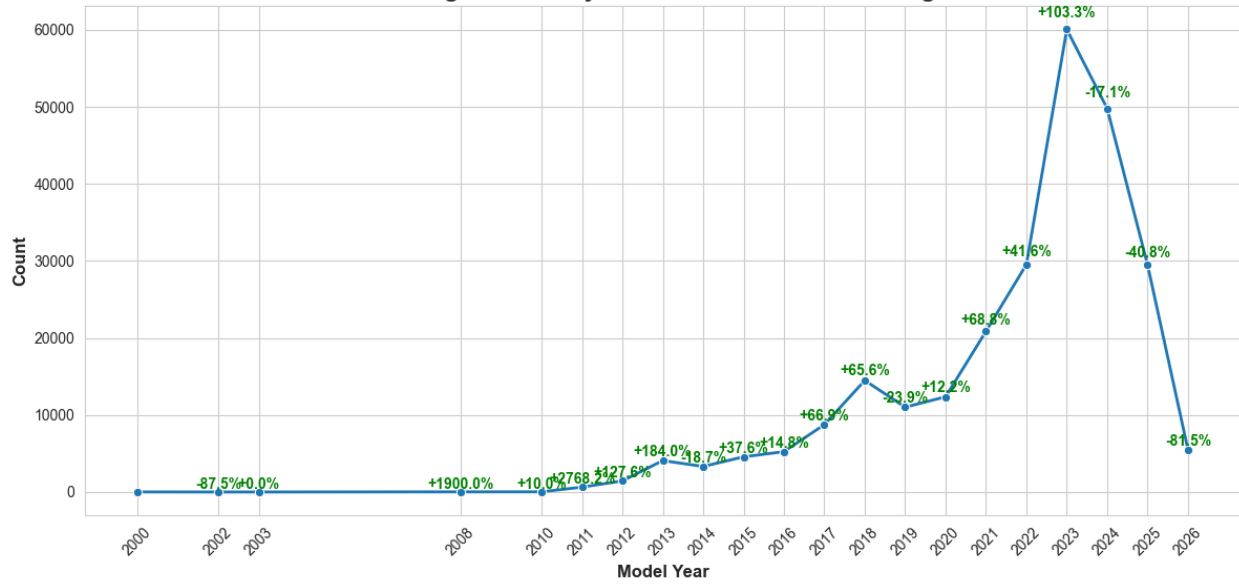
Mass Expansion (2021–2023): Explosive growth (+68.8% in 2021, +103.3% in 2023) → 2023 peaked with ~60,000 registrations, strongest year ever.

Market Cooling (2024–2025): Noticeable slowdown (-17.1% in 2024, -40.8% in 2025), likely driven by supply chain issues, policy shifts, or early adopter saturation.

Partial 2026: Current -81.5% YoY reflects incomplete data, not necessarily a collapse.



EV Registrations by Model Year with YoY % Change



5. Distribution of Electric Ranges by EV Type with Median Comparison

PHEVs (blue curve): Median range: ~25 miles.

Confirms PHEVs are designed for short commutes, after which they switch to gasoline.

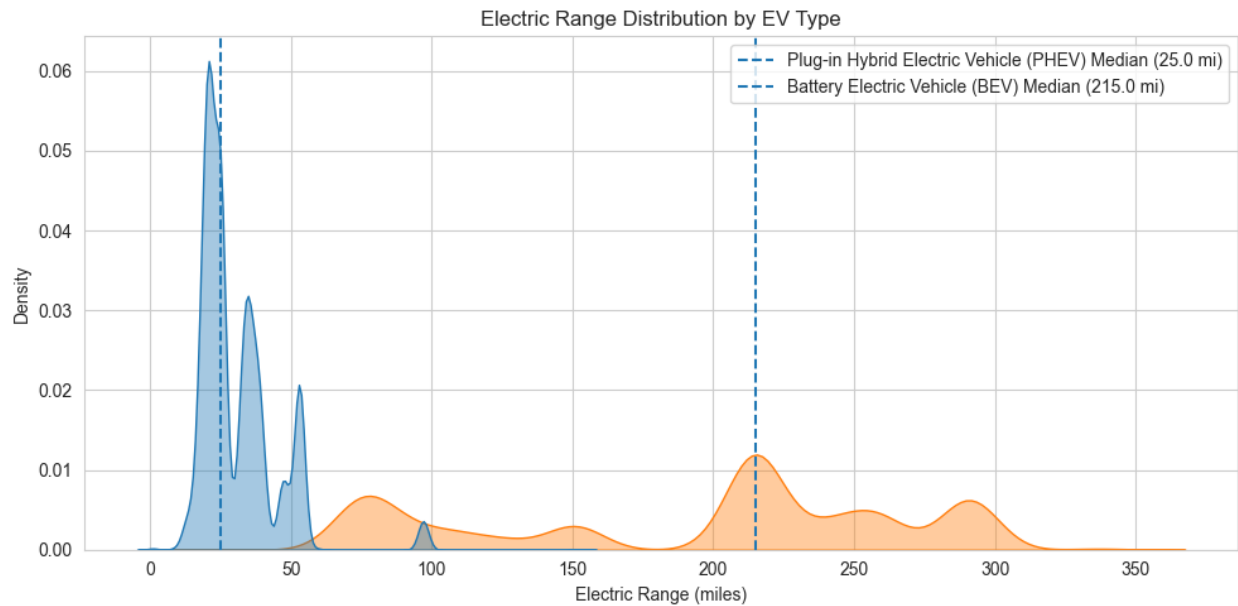
Clustered tightly between 0–50 miles. Indicates reliance on gasoline engines for extended range — electricity covers short urban commutes only. High peaks around 20–30 miles, reflecting design intent for city driving.

BEVs (orange curve): Median range: ~215 miles.

Demonstrates BEVs are viable for longer trips, especially in regions with reliable charging infrastructure. Much wider distribution, ranging from ~70 to over 350 miles. Multiple peaks (100, 200, 250, 300 miles), reflect diverse model offerings from economy EVs to long-range premium models.

Distribution shows clear technological progression in range capabilities.

Median aligns with current industry targets (200+ miles considered acceptable for mainstream adoption).

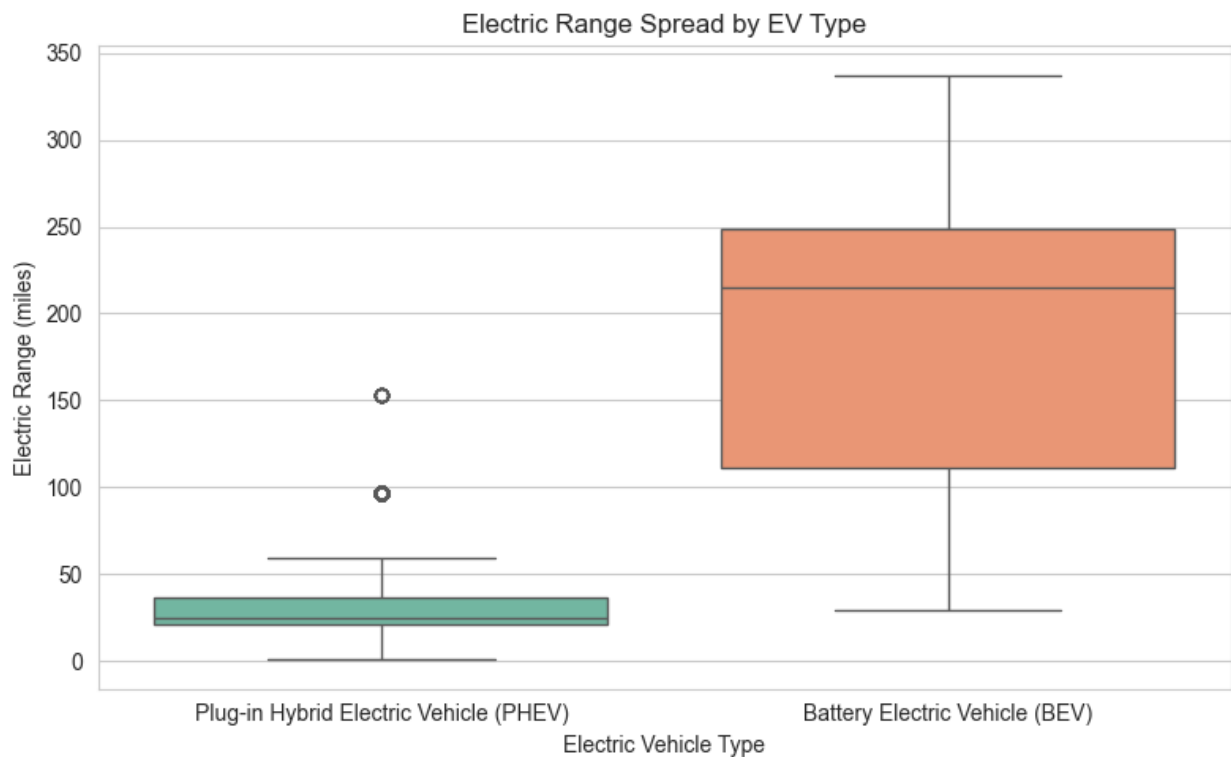


PHEVs (green box):

Very narrow spread, most clustered between 20–50 miles. few outliers (100–150 miles), but generally consistent. Suggests manufacturers design PHEVs for short, consistent urban ranges.

BEVs (orange box):

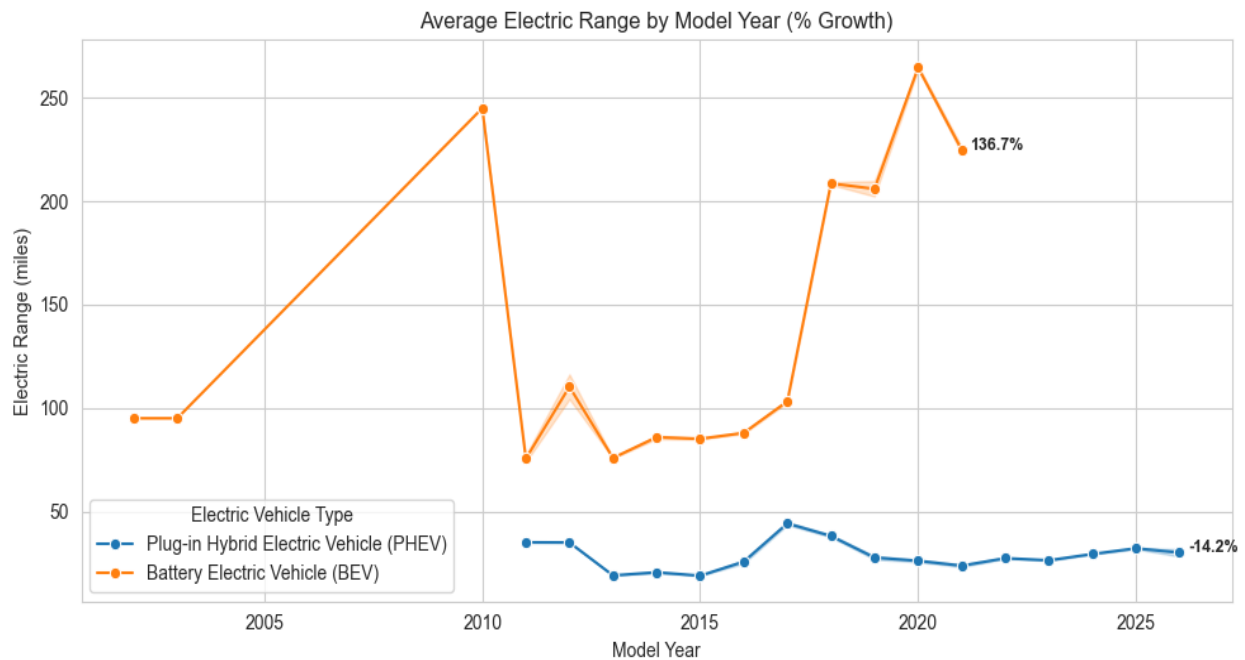
It is widespread ranging from ~30 to 340 miles. Median around 215 miles, but variability reflects different market tiers: Entry-level BEVs: 100–150 miles, Mainstream BEVs: 200–250 miles, Premium BEVs: 300+ miles. Indicates a more mature and diversified market.



6. Average Electric Range by Model Year (% Growth)

Battery Electric Vehicles (BEVs): Showed dramatic growth from ~100 miles (early 2000s) to 260+ miles in 2020s. Total growth over the period: +136.7%, Sharp jumps around 2010 and 2018, reflecting battery innovation and scaling (e.g., Tesla, improved lithium-ion tech). Indicates a mature and rapidly advancing technology.

Plug-in Hybrid Electric Vehicles (PHEVs): Started low (~30–40 miles) and remained almost flat over 20+ years. Growth is negligible and even slightly negative (-14.2%) in recent years. Manufacturers aren't investing heavily in extending PHEV range, treating them more as short-range supplements to gasoline engines.

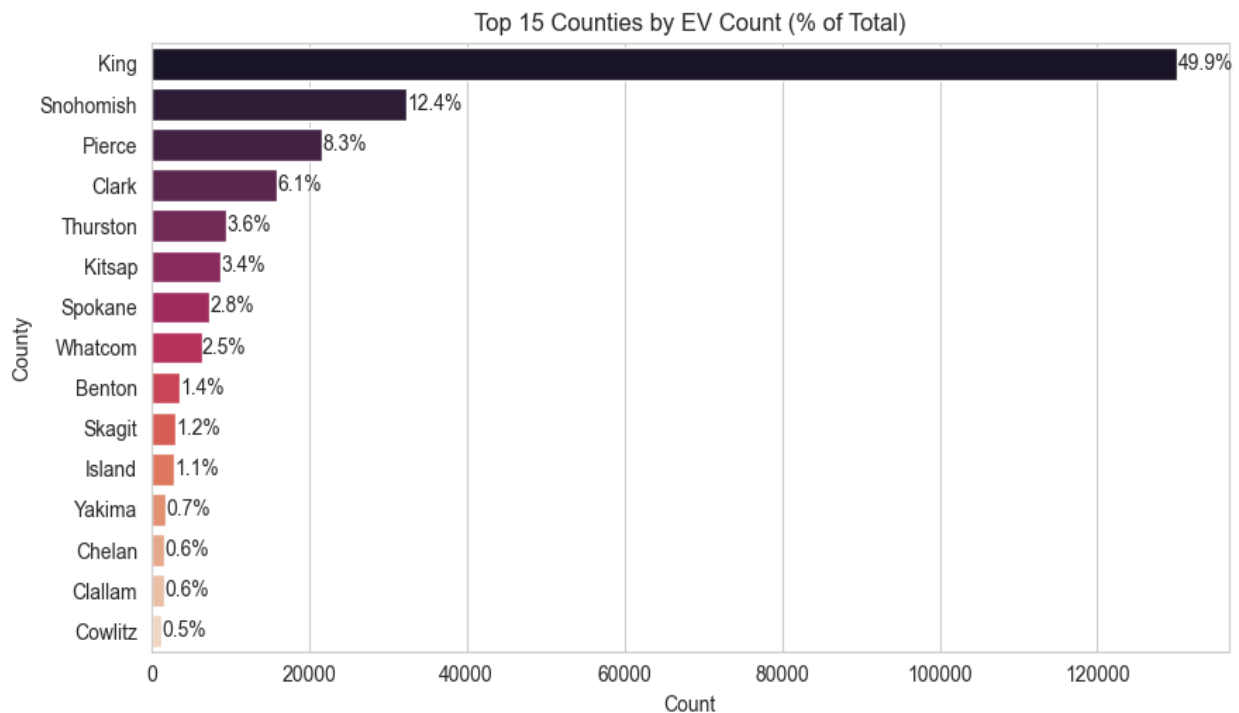


7. Counties by EV Count (% of Total)

King County dominates adoption: Accounts for 49.9% of all EVs in the dataset. Indicates a hub of early adoption, likely driven by higher income levels, strong urban infrastructure, and proximity to major employers (e.g., tech hubs).

Next-tier counties (Snohomish, Pierce, Clark): Combined, these three counties contribute another ~27% of the EV total. Suggests a spillover effect from the urban center (King County).

Remaining counties (<5% each): EV presence is diffuse but growing, with counties like Spokane, Whatcom, and Thurston showing moderate uptake. Smaller, rural counties (Chelan, Clallam, Cowlitz) represent <1% each, highlighting gaps in adoption.



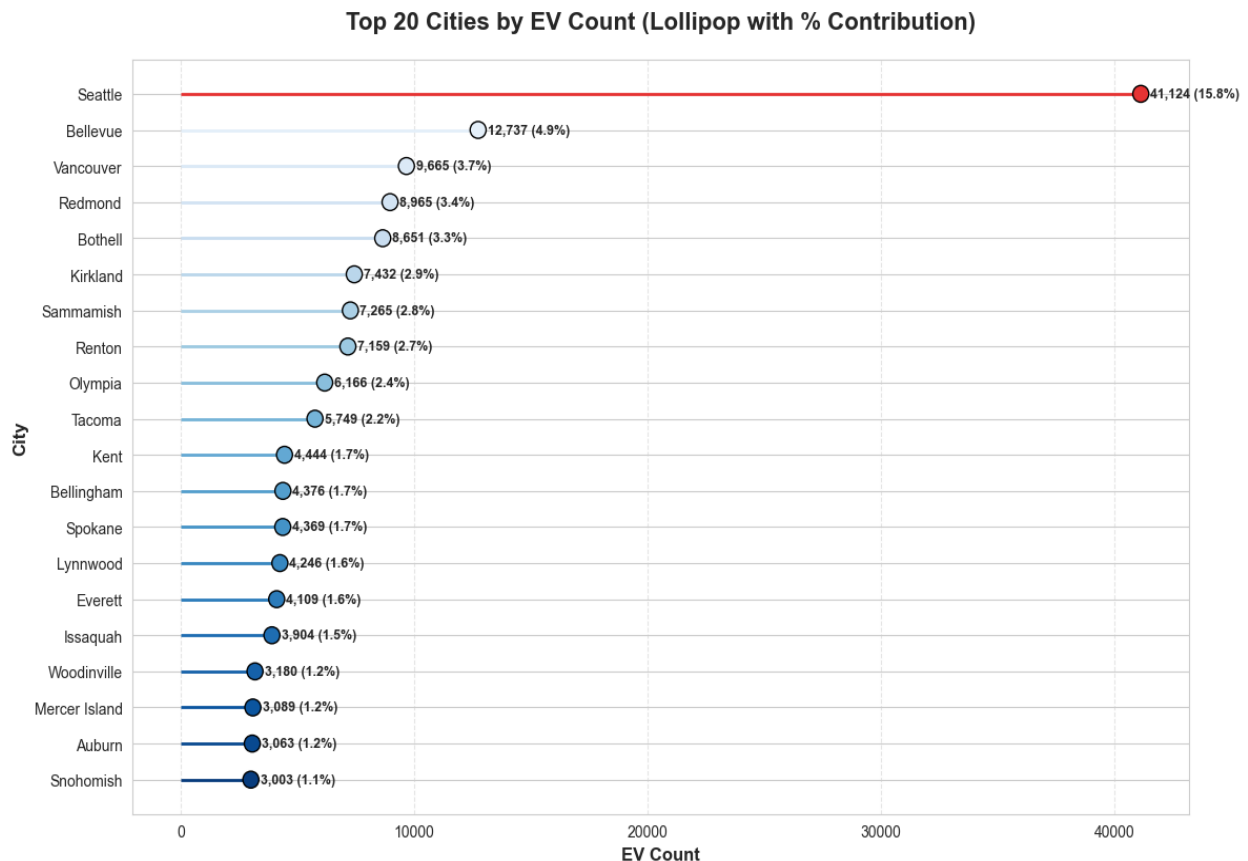
8. Top 20 Cities by EV Count (with % Contribution)

Seattle dominates statewide EV adoption: 41,124 EVs, representing 15.8% of the total fleet. Nearly 3.5× the next-highest city (Bellevue: 4.9%), highlighting Seattle as the primary EV hub.

Strong adoption in major urban/suburban centers: Bellevue (4.9%), Vancouver (3.7%), Redmond (3.4%), and Bothell (3.3%) are the next leaders. Together, the top 5 cities represent ~31% of statewide EVs.

Mid-tier adoption cities (2–3% each): Includes Sammamish, Renton, Olympia, Tacoma. These are growing suburban hubs, indicating regional spillovers from major EV hubs.

Lower-tier cities (<2% each): Examples: Kent, Bellingham, Spokane, Everett, Issaquah. While smaller, their inclusion in the top 20 shows broadening adoption across both urban and suburban regions.



EV Distribution by top 10 + Others

